

Differentially Private Image Obfuscation

Method Section

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This section describes the implementation of two image obfuscation mechanisms based on ε -Differential Privacy (DP): DP-Pixelization and DP-Blur. Both mechanisms use the Laplace mechanism to provide formal privacy guarantees. We evaluate each mechanism across a range of privacy budgets $\varepsilon \in \{0.1, 0.3, 0.5, 0.7, 1, 3, 5\}$ and measure image distortion using MSE and SSIM.

1 Privacy Definition

Given two adjacent datasets D and D' that differ in exactly one record, a randomized mechanism $\mathcal{M}$ satisfies ε -Differential Privacy if for all measurable output sets S :

$$\Pr[\mathcal{M}(D) \in S] \leq e^\varepsilon \cdot \Pr[\mathcal{M}(D') \in S]$$

The parameter $\varepsilon > 0$ controls the privacy-utility trade-off: a smaller ε provides stronger privacy at the cost of greater noise.

2 Laplace Mechanism

For a real-valued query function $f : \mathcal{D} \rightarrow \mathbb{R}$, the ℓ_1 global sensitivity captures the maximum change in f over all pairs of adjacent datasets:

$$\Delta f = \max_{D, D'} \|f(D) - f(D')\|_1$$

The Laplace mechanism adds calibrated noise drawn from the Laplace distribution:

$$\mathcal{M}(D) = f(D) + \text{Lap}\left(\frac{\Delta f}{\varepsilon}\right)$$

This mechanism satisfies ε -DP [1].

3 DP-Pixelization

Mechanism. The image I of size $H \times W$ is partitioned into non-overlapping $k \times k$ blocks. For each block, the mean pixel value is computed and independent Laplace noise is added:

$$\tilde{p}_{\text{block}} = \frac{1}{k^2} \sum_{i=1}^{k^2} p_i + \text{Lap}\left(\frac{\Delta f_{\text{pix}}}{\varepsilon}\right)$$

The noisy mean is then assigned uniformly to all pixels within the block, and the result is clipped to $[0, 255]$.

Sensitivity Derivation. Pixel values lie in $[0, 255]$. In the worst case over adjacent datasets, a single record change causes all k^2 pixels within a block to shift from 0 to 255 (or vice versa). The resulting change in the block mean is:

$$\Delta f_{\text{pix}} = \frac{1}{k^2} \cdot (255 - 0) = \frac{255}{k^2}$$

A larger block size k reduces the sensitivity, and therefore requires less noise to achieve the same ε . However, the block-averaging operation itself already introduces geometric distortion as an additional source of visual protection. We experiment with $k \in \{8, 16\}$.

4 DP-Blur

Mechanism. A Gaussian blur is first applied to the image:

$$B(x, y) = \sum_{(i, j)} G_{\sigma}(i, j) \cdot I(x + i, y + j)$$

where G_{σ} is a Gaussian kernel with standard deviation σ . Laplace noise is then added independently to each pixel:

$$\tilde{p}(x, y) = B(x, y) + \text{Lap}\left(\frac{\Delta f_{\text{blur}}}{\varepsilon}\right)$$

Sensitivity Derivation. Gaussian blur is a linear operation whose output at each location is a weighted average of input pixels, with kernel weights $w_i \geq 0$ and $\sum_i w_i = 1$. Taking the conservative per-pixel upper bound, the maximum possible change in any single output pixel is:

$$\Delta f_{\text{blur}} = 255$$

We use $\sigma = 1.0$ with a 5×5 kernel in all experiments.

5 Evaluation Metrics

For each value of ε , we compute the following metrics between the original image I and the obfuscated image $\tilde{I}$, averaged over all 400 images in the AT&T (ORL) face dataset:

- **Mean Squared Error (MSE):**

$$\text{MSE} = \frac{1}{HW} \sum_{x=1}^H \sum_{y=1}^W (I(x, y) - \tilde{I}(x, y))^2$$

A higher MSE indicates greater pixel-level distortion.

- **Structural Similarity Index (SSIM) [2]:** SSIM jointly evaluates luminance, contrast, and structural information, producing a score in $[-1, 1]$ where 1 indicates perfect similarity. Unlike MSE, SSIM correlates better with human perception of image quality.

As ε increases, the privacy guarantee weakens, less noise is injected, and we expect MSE to decrease while SSIM increases. These curves are reported in Figure ??.

6 Privacy–Utility Trade-off

The Laplace noise scale $b = \Delta f / \varepsilon$ governs the trade-off:

Mechanism	Noise scale b	Sensitivity
DP-Pix ($k = 8$)	$255/(64\varepsilon)$	$255/64 \approx 3.98$
DP-Pix ($k = 16$)	$255/(256\varepsilon)$	$255/256 \approx 1.00$
DP-Blur	$255/\varepsilon$	255

DP-Blur injects substantially more noise than DP-Pixelization at the same ε , because each pixel is treated as an independent query with full sensitivity 255. DP-Pixelization amortizes the sensitivity over k^2 pixels per block, resulting in a lower noise requirement.

References

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